Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/30/2006
PI Name:	Leary, James F Ph.D.		
Project Title:	Nanoparticle Delivery of Repair Enzymes for Radiation Protection/DNA Repair		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907	**Congressional District:**	4
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/30/2001	**End Date:**	12/20/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Motamedi, Massoud (University of Texas Medical Branch) Kotov, Nicholas (University of Michigan) Lloyd, Robert (Oregon Health Sciences Center) Lvov, Yuri (Louisiana Tech University)		
Grant/Contract No.:	NAS2-02059		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This program will develop a nanotechnology-based biosensor diagnostic and therapeutic system to detect potentially radiation-damaged cells and to provide radiation protection and/or DNA repair enzymes to white blood cells in the peripheral blood of astronauts exposed to radiation in space. A multidisciplinary team of scientists with expertise in targeting and detection of rare blood cells, DNA repair, nanoparticle technology, and in-vivo imaging will develop a minimally invasive, smart-biosensor technology that can be potentially read through the blood vessels in the eyes of astronauts. The focus of this grant is to provide as much in-vivo protection against radiation damage to the blood and bone marrow of astronauts, and to provide in-vivo intra-cellular DNA repair to damaged cells before these cells go on to become radiation- induced leukemias. The work will be performed by five interacting groups with the PI performing the overall integration for this high-risk, high-payoff, multidisciplinary research. We have attempted to address key features of the NASA-NCI collaboration as outlined in BAA N01-CO-17016-32 Fundamental Technologies for Development of Biomolecular Sensors. The overall objective is to develop “nanomedicine”, integrated smart nanomedical systems for simultaneous diagnosis and therapy of disease, treating one cell-at-a-time – essentially nanosurgery inside single living cells to attempt to repair, if possible, injured single cells. These nanomedical systems will have applicability to a wide range of human diseases. We address what is described as a key problem of interest to NASA, namely a novel way to deal with radiation protection and repair of human cells in-vivo in response to ionizing radiation doses in space that are currently impossible to shield against without impossible launch mass requirements (Task 1). New chemistries and nanostructures will be developed to produce novel, targetable multi-layered, biopolymer coated nanoparticles that can be used for both in-vivo biomonitoring and therapeutics of space radiation exposures and also suitable for ground-based protection and repair of normal cells in cancer patients undergoing radiation therapy (Task 2). The biocompatibility of nanomaterials, efficiency and accuracy of targeting, the development of molecular biosensors, and delivery of genes/drugs is being integrated into nanosystems incorporating these features (Task 3). The use of novel imaging technologies as part of our in-vivo assessments of the biocompatibility, targeting and therapeutic effectiveness of our nanosystems tested on animals in-vivo (Task 4). We believe this proposal addresses the technological objectives of the BAA by developing a new generation of in-vivo, biomolecular sensor/therapeutic nanotechnology to deal with the critical problem of prevention, direct intracellular detection of radiation damage, and repair, if possible, to blood cells of astronauts that have received unpreventable radiation exposure in long duration space flight. This work also has importance to an important problem on Earth, namely the protection and recovery of normal cells injured during radiation therapy of cancer patients. The concept of attempting to repair living cells one cell-at-a-time would represent, if successful, a significant advance in medicine, building on the unique feature of nanotechnological repair tools the size of the damaged molecules they are designed to repair. A significant challenge to the implementation of manned, long-duration space travel and exploration is the maintenance of human health in deep space. In addition to the complex issues associated with health maintenance in a zero gravity environment, such as weakening of the immune system to deal with infectious diseases and cancers, humans will be exposed to high levels of ionizing radiation (IR), that cause a variety of DNA damages. If inadequately repaired, these DNA lesions will be converted into permanent mutations that ultimately can result in a variety of disease states, including cancer. This proposal identifies methodologies and strategies that will not only detect the major IR-induced DNA base lesions that are known to be highly mutagenic, but also address issues of radiation protection and prevention of this damage by engineering human cells to become more resistant to the damaging effects of IR using the addition of new DNA repair genes into progenitor cells. Thus this proposed work involves the biosensing and targeted delivery of protective and repairing biomolecules to radiation-exposed blood cells for early detection of cells that would otherwise either be killed or transformed into malignant cancer cells. The purpose of this very early detection of radiation-exposed cells is to reduce the level of cell damage and the probability of developing radiation-induced cancers in space or on the ground.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall objective is to develop “nanomedicine”, integrated smart nanomedical systems for simultaneous diagnosis and therapy of disease, treating one cell-at-a-time – essentially nanosurgery inside single living cells to attempt to repair, if possible, injured single cells. These nanomedical systems will have applicability to a wide range of human diseases.
Task Progress:	Year 1: 1. Performed calculations and constructed proteins for use in DNA binding/repair process. 2. Assembled and tested semiconductor and polyanion nanoparticles for project use. 3. Tested nanoparticle binding and targeting to specific cell subpopulations. 4. Tested in-vivo biocompatibility and biodistribution of nanoparticles. Year 2: 1. Characterized DNA repair capacity of wild-type and repair-deficient human cells. Optimize repair enzymes using computational approaches to structure/function relationships. 2. Developed biocompatible coatings for nanoparticles. Develop multi-layered nanoparticles for targeting and delivery of repair enzymes. 3. Provided single-cell/coated nanoparticle biocompatibility data and assessment of targeting specificity and efficiency. 4. Developed capability for in vivo non-invasive detection and imaging of coated and targeted nanoparticles to specific cells in animal models. Year 3: 1. Further elucidated the molecular mechanisms of radiation injury and repair occurring in cells. 2. Optimized nanoparticle targeting and repair enzyme loading procedures. 3. Assessed nanoparticle-targeted cell complexes for evidence of DNA repair in these cells. 4. Developed "nanoparticle factories" for intracellular production of therapeutic genes under molecular biosensor control. 5. Developed optical contrast agents for in vivo imaging and sensing.
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Patents	Full US Patent PCT Application PCT/US05/06692. Filed March 4, 2005. Mar-2005 Leary JF, Prow TW. "Multilayered Nanomedicine Delivery System and Method."